

Python For Data Science

Total Time: 3 Hours

Theme: Healthcare Predictive Analytics Foundations

Data Set: <https://www.kaggle.com/datasets/ronitf/heart-disease-uci>

Section A: Concept Application

1. You open the Heart Disease dataset in Jupyter Notebook. What is the difference between a .py file and a .ipynb file?
2. The dataset has columns like age, cholesterol, and sex. What is the difference between a numerical feature and a categorical feature?
3. You run `type()` on the target column and it returns `int`. Why does the data type of your target variable matter before framing a prediction problem?
4. You compute the correlation between age and heart disease presence and get 0.23. What does a correlation value tell you about two variables?
5. A hospital rule says: if cholesterol > 240, flag as high risk. What is the fundamental difference between a rule-based system and a data-driven approach?

Section B: Practical Task

1. Implement an automated data ingestion pipeline using Pandas to load the UCI Heart Disease dataset and perform type-casting for categorical variables like 'sex', 'cp', and 'restecg'.
2. Execute a comprehensive outlier detection strategy on 'chol' (cholesterol) and 'thalach' (max heart rate) using the Interquartile Range (IQR) method and visualize the results via Boxplots.
3. Conduct multivariate analysis using Seaborn Heatmaps to identify multi-collinearity between clinical features and determine which variables hold the strongest correlation with the target 'heart disease' label.

4. Develop a feature engineering script to normalize numerical data using StandardScaler and create derived features such as 'Age-adjusted Max Heart Rate' to improve predictive signal.

Section C: Mini Project

1. **Title:** Cardiac Risk Stratification and Clinical Decision Support System
2. **Problem Statement:** Develop a diagnostic analytical framework to predict the presence of heart disease, allowing clinicians to prioritize high-risk patients and transition from reactive care to proactive preventive intervention.
3. **Dataset Recommendation:** Heart Disease Dataset (UCI / Kaggle)
<https://www.kaggle.com/datasets/ronitf/heart-disease-uci>
4. **Required Deliverables:** Cleaned and pre-processed CSV file, Jupyter Notebook with documented EDA, Feature-Target interaction matrix, Comparison report on "Data Science Predictive Modeling vs. Static Rule-Based Thresholds."